RESEARCH REPORT
GIA LAI ELECTRICITY JSC (HSX.GEG)

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GIA LAI ELECTRICITY JSC (HSX.GEG)

Ticker: GEG
Sector: Utilities
Industry: Renewable Electricity Producers

Date: Dec.15, 2021Target price: VND 22,700
Recommendation:BUY
Upside: 11.9%

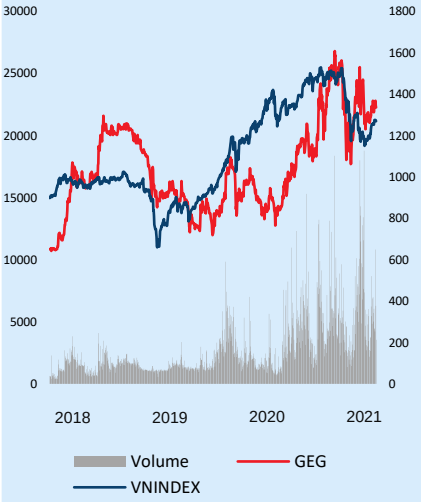
Recommendation: BUY

MARKET SNAPSHOT	
Market Cap (VND bn)	5,937
Share outstanding (mm)	303,715
15-day average volume	720,000
52-week high (VND)	22,200
52-week low (VND)	13,512
Beta	0.74
Free float (%)	30%
Expected dividend 2022 (cash)	0%

Key firgures

	2018	2019	2020
EPS (VND/sh)	1.254	1.06	1.029
P/E	12,16	21,3	16,67
P/B	1,2	2,2	1,6
ROE (%)	10,1	11,6	9,8
ROA (%)	6,2	5,2	4,1
D/E	0,81x	1,62x	1,24x

Price history GEG vs. VN-Index rebased



GEG: GONE WITH THE WIND

EXECUTIVE SUMMARY

I propose a **BUY** recommendation for GEG with a one-year target price of VND 22710, presenting a 12% upside potential on the closing price of VND 22,800 on December 15th 2021.

Macroeconomic stability reinforces GEG's industry position

The recent movement in Vietnam's towards higher living energy policy standards and rapid urbanization consolidates to growing strategy of Gia Lai Industry in the green energy market. In 9/2021, total electricity consumption has increase at a staggering rate of 133% since the reopen of industrial park and production. Along with Vietnam economy's piecemeal retrieval from the Covid-19 bombard, continuous innovations have also contributed to GEG's better adaptation to economical hardship and progressively demanding market requirements. For these reasons, we believe GEG's expanding strategy in offshore wind energy makes it an indirect beneficiary of Vietnam GDP growth, driving a slight fall but still resilient CAGR in domestic sales 2020F-2024F while expanding margins (13.2% 2022F to 20% 2026F).

Power Development Plan VIII supports GEG to expand its wind energy plants

The global renewable energy Industry impressively grew in 2021. The additional renewable energy capacity has been set to rise by nearly 290 GW, an increase of 3% growth rate YoY. In which, Solar Power accounted for 50% of the total capacity, followed by Wind Power and Hydropower. It's forecasted that the world will host 4,800 GW of the global renewable energy capacity by 2026 and rocket by more than 60% YoY, equivalent to the current global capacity of Fossil Fuels and Nuclear. In addition, the world government governments are moving toward a greener and less pollutive energy policy by committing on COP26 Agreement. To implement COP 26 commitments, the MOIT is urgently finalizing the VIII National Power Development Plan under the Prime Minister's instruction to send it to Local authorities for discussion before finalizing and reporting to the Prime Minister in Q2/2022.

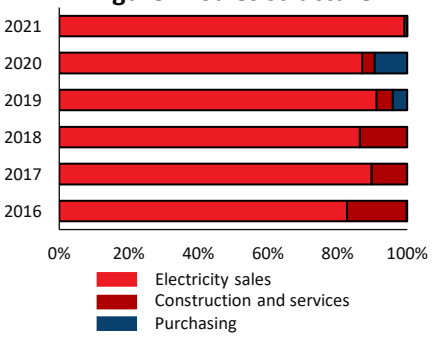
Competent ability and effective business model promise a resilient gain

GEG's competitive advantages lie in its vertically integrated value chain and technology advantages. Wind power is the key growth driver enabling a robust growth of core business, recorded in 100% increase of GEG's energy output. GEG's technology will play a pivotal role in the GEG value chain by smoothing out intermittent generation from wind and solar energy. Adjusting for the negative after effect of Covid-19, this ambitious plan would build a strong foundation for future development of business operation through the expectation of revenue achieving a CAGR of 23% in the period of 2022-2026F as our team's estimation.

Despite these stimulating factors, GEG is facing a relatively high financial distress

To compromise with the company strategy in expanding solar and wind energy, GEG's liabilities is increasing aggressively in recent year. In 2021, its Debt-to-Equity ratio is 2.9, higher than the median of the industry-1.5. Although the investment rate of solar power of GEG is comparably lower than the average of the industry, but the raise in wind turbine and steel price have contributed to the debt payment pressure. In addition, the tightening of Vietnam's policy on bond issuance is making it more difficult for GEG to access in one essential source of fund to bolster V.P.L 1-wind power turbine.

Figure 1: Sales structure



Source: GEG

Figure 2: Ownership structure

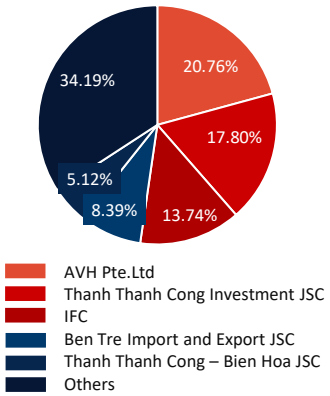
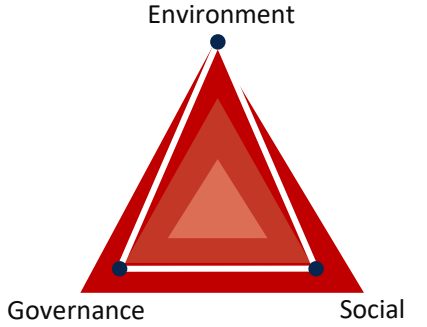
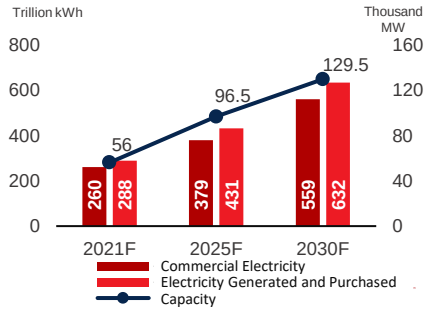


Figure 3: ESG



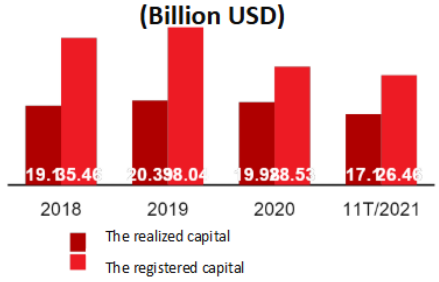
Source: Company data

Figure 4: Electricity sector output and capacity forecast



Source: Refinitv

Figure 5: Vietnam annual FDI inflows (Billion USD)



Source: MPI

Business description

GEC is a subsidiary of TTC Group - a multidisciplinary private corporation operating in the fields of Agriculture, Energy, Real Estate, Hospitality and Industrial Real Estate with total assets of USD3.5 billion. From a State-owned enterprise with only 1.2 MW of Hydropower, GEC, which has 33 years of establishment and continuous development, has been equitized, strongly grown and become the sole enterprise listed on the stock market operating Renewable Energy from Hydropower, Solar Power, Wind Power with a total portfolio of nearly 700 MWp by the end of 2021. With the target of 1,700+ MWp total capacity by 2025, GEC aims to expand into new types of Renewable Energy such as Waste-to-Energy and Biomass.

Company strategy

Broaden product portfolio

GEG is the pioneer in providing renewable energy; the company is embarking an ambitious plan on expanding its current capacity-545 MWh to 1700 MWh in 2025 (which is 3 times higher than its current capacity) by aggressively developing large power plants. In addition, to capture the rising market demand and keep up with fast-changing of modern life, GEG is targeting to penetrate and conquer the ample market share of alternative renewable energy such as: LNG power, Biomass,… before the advent of international player and other renewable energy brand.

Engineering-focus to utilize power plant maintenance cost

Acknowledging the crucial role of an integrated value chain, GEG is re-positioning its engineering service to approach other energy power plant company in Vietnam. This segment is considered one of the fastest growing segment in the renewable energy due to the following factors: (1) the number of wind power plant is surging, (2) nuclear, biomass, LNG power development are potential but inexperienced, hence an experienced engineer provider is requested. Furthermore, one of GEG current weakness is that its hydropower plant is underperformed. The development of the engineering team will stimulate the O&M process, maximizing its energy output.

ESG driven localized strategy

Besides investing in green energy project, GEG is continously increasing the investment capital in labor resources, local habitat conservation and the environmental protection plan. This strategy will compromise with not only the company strategy to bring a robust and sustainable business environment but also the government guideline and policy to cope with its target of zero carbon emission by 2050.

INDUSTRY OVERVIEW

Energy sector is exploding

Due to the low labor, low industrial lending cost, Vietnam is one of the best destination besides China for big corporate to inject their FDI. In 2022, Apple suppliers Luxshare Precision Industry and Foxconn have started test production of the Apple Watch in northern Vietnam with the aim of producing the device outside of China. The flow of fund into alternative energy is also increasing. The total value of investment in renewable energy has climbed as an astonishing rate of 32% CAGR 2012-2022. These factors are the key macroeconomic driver to the development of the energy sector in Vietnam. In 2021, Vietnam produced 255 TWh of electricity output, recorded as a staggering 7% CAGR for the last 5 years-which is higher than that of GDP growth.

Favourable legal framework and infrastructure expansion pave the way for renewable energy

The new power purchasing mechanism-Direct Power Purchasing Agreement (DPPA) with a highly decentralized distribution system issued by EVN will shed the light for firms to increase their revenue and competitiveness. The MOIT and the related Ministries have commented on VIII Power Development Plan according to the Prime Minister's commitment to reduce emission to Zero - Net Zero at the United Nations Climate Change Conference - COP 26. Accordingly, to minimize Coal Thermal Power

Figure 6: Porter's Five Force analysis

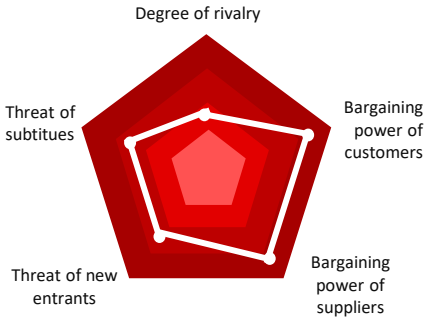
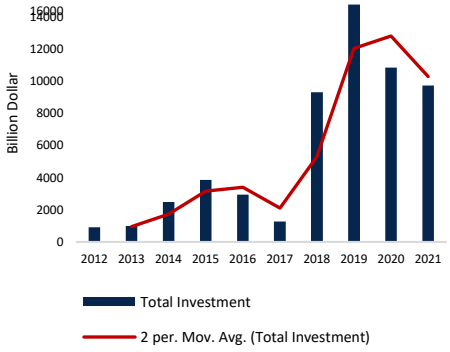
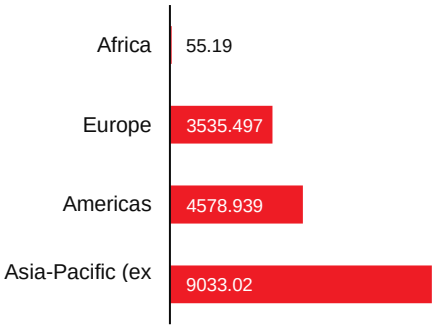


Figure 7: Total PE investment in RE sector in Asia in 2012-2021



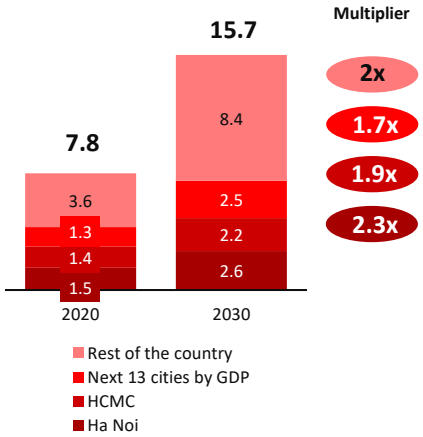
Source: Refinitiv

Figure 8: Gross Proceed from IPO in RE sector across 4 regions (Million Dollar)



Source: Refinitiv

Figure 9: Household with annual income greater than $22,000



Source: McKinsey

Plants and the ratio of RE sources outside of Hydropower Plants has been increased by more than 30% in 2030 and 44% in 2045. In addition, the VIII Power Plan also suggested to ameliorate the 500Kv transmission line to improve the quality of electricity and cut down on loss during conveying power to rural or high-demand area such as: Southeast, Mekong Delta, Central Highlands, Northwest,… To further stimulating the sector, the government issued Decree 218/2013/NĐ-CP to cut down on corporate tax on the renewable energy sectors-the cutting rate fluctuating from 10% to 50% depending on the size and construction time of the project.

COMPETITIVE POSITIONING

GEG enjoys a 20-year of higher FIT price setting with EVN

Thanks to the professionalism of GEG and IPC construction, all 3 of GEG wind power plants came into commercial operation before the FIT extension period expires. This not only made the future cashflow of GEG to be stable but also outperformed of its rivals at a margin of 20%-30% (due to the FIT price is higher than the PPA that EVN signed with other RE firms).

Collaboration of renowned foundation in the RE industry

GEG is continuously being supported from foreign shareholders such as: JERA-Tokyo Electric Power and Chubu Electric Power, IFC, Armstrong,… in the financial and professional aspects. Additionally, Vestas- a distinguished foundation that produce wind turbine and offer advisory service for offshore wind plant is the main suppliers for GEG wind power plant. Recently, GEG signed with Vestas to supply wind turbines and O&M services in 10 years with a system of commitment to use quantitative method to ensure maximum operational efficiency of the project.

Self-operated and vertically integrated RE value chain

GEG is continuing to complete and put into operation a closed value chain in the RE segment to minimize the impact of the market, diversify revenue sources, ensure sustainable growth of business targets and utilization of costs.

Porter 5 Forces (**Appendix C.2**)

INVESTMENT SUMMARY

I propose a BUY recommendation on Gia Lai Electricity JSC (GEG) with a target price of VND 22,750 using the Discounted Free Cash Flow to Firm and Relative Multiple Valuation method. This presents a 12.1% upside from its closing price of VND 20,290 on Dec 15th, 2021 and implies a 22x 2022F P/E.

Buoyant disposal income fosters energy demand-specifically RE segment

GEG is the racing horse in Vietnam's energy industry, reaching out for being an iconic pioneer in Renewable energy development in Vietnam. This is supported by a sharp increase in household disposable income **(Figure 9)**, driving margin expansion. This ambition is no longer too far to achieve as three recent years recorded a promising situation as Vietnam's renewable electricity consumption and input installed capacity is consecutively increasing at a rate of 7%-despite the outbreak of Covid-19, convincing its stable and growing demand for renewable electricity among small Asian markets while the rest of the world was in decline. The RE industry, hence, is an attractive field for new players-specifically foreign investors to join in, but in my view, GEG's positioning allows it to indirectly benefit from domestic consumption growth while enjoying limited threat of substitutes in local businesses. I believe GEG is well-positioned to capture rising consumer spending in the Vietnam.

Outstanding performance amd effective business model assure GEG as a professional RE investors

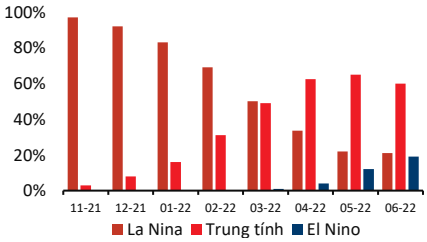
Vietcombank evaluates each project for its efficacy outcomes, the possibilities for partnering with other parties to carry out the project, and the approval of the owner's 70% credit limit. In the past, many homes have received credit support for the enormous renewable energy machine from GEG. Vietcombank's Krong Plant Pa

Figure 10: Projects financed by Vietcombank

Project	Total investment (Billion VND)	Credit ratio	Capacity (Qr/Q)
Solar power			
Krong Pa - Gia Lai	1301	67%	100%
Ham Phu 2	979	74%	120%
Duc Hue 1	936	68%	120%
Wind power			
Ia Bang	1955	70%	N/A
Tan Phu Dong 2	2242	70%	N/A
VPL 1	1320	70%	N/A

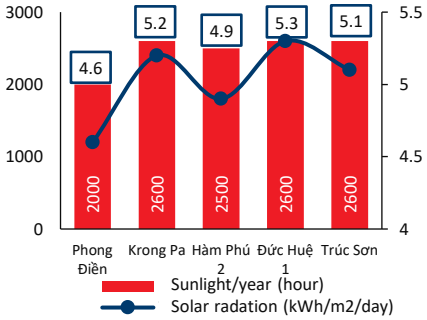
Source: GEG

Figure 11: ENSO occurrence forecast



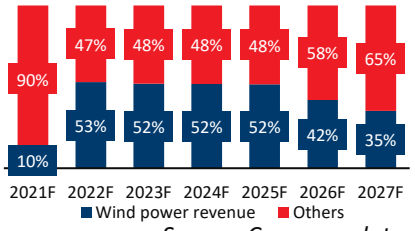
Source: NOAA

Figure 12: Solar plants' output and radiation



Source: Company data

Figure 13: Wind power revenue proportion



Source: Company data

Solar Power in Gia Lai, Ham Phu 2 in Binh Thuan, and Duc Hue 1 in Long An are the most well-known applications (**Figure 10**). These projects operated with real production exceeding the plan and were finished ahead of schedule.

Supportive investment climax invigorating Renewable energy sector

Hydroelectric energy

Vietnam is largely a tropical monsoon country. In the north, a hot rainy season prevails from May to September. In the south, Ho Chi Minh City and the Mekong Delta experience a year-round tropical climate with daily temperatures normally exceeding 88°F. The rainy season in Danang and Hue in the center of the country lasts from October to March. With more than 3500 rivers and 2500 millimeters of rainfall per year, Vietnam has the second largest hydroelectricity in SEA region. Furthermore, the probability of La Nina to return is increasing, thus fertilize hydroelectric dams.

Solar power

Lower initial CAPEX and vertically integrated value chain directly benefit GEG solar power revenue. Since its first launched in 2018, solar power has been the core business besides hydroelectric and contributed most to the revenue structure-It contributed 58% of revenue structure and exhibited a stable 10% YoY growth. Moreover, the radiation volume from 4.6-5.3 Kwh/m2/radiation and sunshine hours from 2000-2520 hours/year (higher than average from 7-8%) in Vietnam will give GEG stable output. In addition to favorable natural conditions, factories also have locations close to the existing high voltage (110kV) transmission line, which reduce investment costs and capacity cutdown by EVN. (**Figure 12**)

Wind-power

Vietnam has a favorable natural potential for wind capacity with a long coastline of 3,000 km and winds that blow from 5.5 to 7.3 meters per second. By the end of 2020, Vietnam's installed wind power capacity reached 600 MW, behind only Thailand among ASEAN countries. The country's offshore wind potential generated from the wind at sea is much larger than onshore wind. According to the World Bank Offshore Wind Roadmap for Vietnam, Vietnam can witness an increase in offshore wind power capacity to 5 – 19 GW from 1 GW and in onshore wind power capacity to 17.34 GW from 1.26 GW by 2030. This will likely generate around US$60 billion in gross value added (GVA) for the country.

Wind-energy focus strategy promise stable and growing margin

In October 2021, GEG officially started commercial operation of 3 wind power plant with a total capacity of 130 MW as planned to enjoy preferential price FiT1 (8.5 cents/kWh applied to projects wind power onshore and 9.8 cents/kWh offshore) for 20 years. These projects will bring GEG of 416 million kWh of electricity and VND 889 billion in annual revenue, accounting for 42% of the total revenue of GEG. (**Figure 13**) It is expected that these 3 plants will contribute to reducing emissions 361,411 tons of CO2/year, bringing the total reduction in emissions of the energy portfolio the company's renewable energy to nearly 800,000 tons of CO2/year. In addition, Tan Phu Dong 1 are implemented to operate in the late 2022, it will generate more than 816 million kWh of production volume and more than 1700 billion annual sales. Consequently, wind power will contribute the largest revenue proportion (about more than 50%) and bolster up GEG's total electricity sales of nearly 3,300 billion - 2.5 times compared to the present.

Figure 14: Energy intensity and energy demand per capita in selected regions in the Announced Pledges Scenario 2020-2030

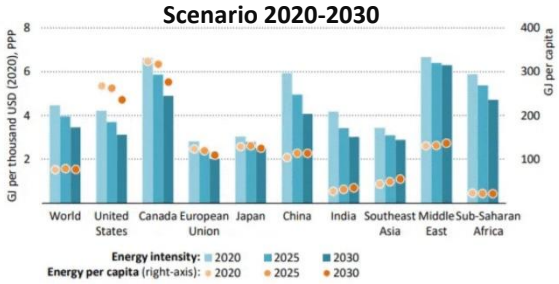
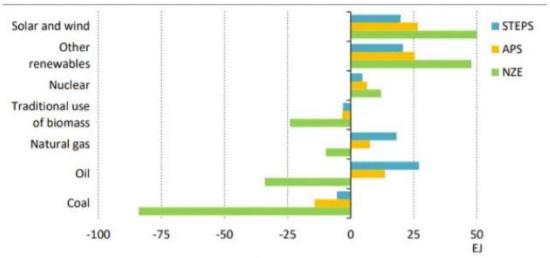
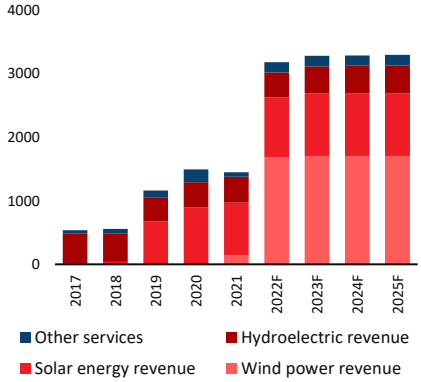


Figure 15: Change in global total energy supply by fuel and scenario 2020-2030



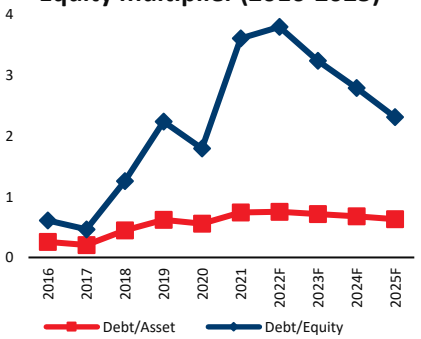
Source: IEA, Energy Report 2021

Figure 16: GEG's revenue projection (Billion VND)



Source: My analysis

Figure 17: GEG's Debt/Asset & Equity multiplier (2016-2025)



Source: My analysis

Figure 18: WACC Calculation

WACC Calculation	
Beta	0.74
Risk free rate	3.70%
Equity Risk Premium	8.21%
D/(D+E)	57.55%
Marginal tax rate	7%
Cost of equity	9.81%
Cost of debt	9.00%
WACC	8.79%

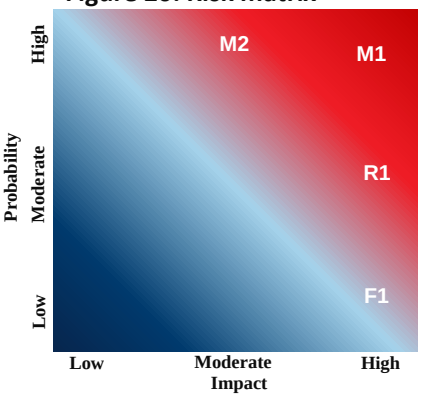
Source: My analysis

Figure 19: Valuation summary

Method	Weight	Price
FCFF	60%	24,980
P/E	40%	19,000
Target price		**22,710**
Current price		20,280
Change		12%

Source: My analysis

Figure 20: Risk matrix



Source: My analysis

FINANCIAL ANALYSIS

One-off event hinders the consecutive growth

On the back of the headroom for growth, GEG has delivered a concrete revenue growth of 32% over the 2016-2021 period. But in 2022, Given the economic downtrend, Renewable energy is considered a spotlight for growth of GEG as it arises as a non-cyclical industry. Meanwhile, Solar power sector is believed to recover in the new normal landscape through the improvement of electric power transmission line.

Relatively weak financial health due to rapid expansion

High debt-to-asset ratio at roughly 0.7x threaten GEG financial position to embrace its expansion and diversification strategy. Moreover, the interest coverage ratio has been plummeting since 2018-the year GEG launched its first solar power plant.

VALUATION

I have considered two typical valuation methods to evaluate GEG, including Discounted Cash Flow (DCF) and Multiple Analysis. While the former method was used to estimate the intrinsic value of share price due to the predictability of cash flows in relation to growth and profitability, the latter one is simply an expression of market value relative to a key statistic.

DCF Valuation: Free Cash Flow to Firm (FCFF)

I evaluates GEG using Free Cash Flow to Firm (FCFF) as the company has unstable capital structure over last years. This method consists of a two-stage growth model. The first phase is based on a specific year to year forecast up to 2027 and the second phase of a constant growth of 1.5% (**Appendix D.4**). According to our detailed analysis, I arrive at the price of VND 29,670 per common stock of GEG.

Relative valuation

Since the lack of domestic listed peers, I compared GEG to Renewable energy companies in APAC regional. As the strategy of the focusing on the Wind power sector, in particular account of 60% revenue in 2023F, I choose the regional peers based on this scope of products (**Appendix D.7**).

INVESTMENT RISK

[M1] Market risk | Weather condition fluctuations waver net income

Variation in the probability of La Nina and El Nino events directly impact the electricity output of GEG. In addition to the former matter, storm and flood will contribute to the output cutdown and increase in maintenance cost of solar panel. According to Brussel Times, in February 2022, all of Belgium Offshore Wind Power Plants were shut downed due to the Eunice Storm. This shutdown resulted in a 50% drop in capacity when the turbine ran again.

[M2] Market risk | Over-investment in wind power project brings about a big cut in output

In recent years, many firm have invested massively in renewable energy created to enjoy preferential electricity prices. According to EVN, in early 2021, the output from renewable energy sources reached 7.79 billion kWh, increased by 180.6% over the same period in 2020.

[R1] Regulatory risk

GEG operates under the influence of the Vietnamese legal system, regulations and policies related. Changes related to regulations, investment process of wind power and price mechanism, bidding mechanism for solar power development affect GEG's investment activities. Policies are also subject to frequent changes in response to actual demand, causing instability of the policy and investors' behavior.

[F1] Financial risk

Due to decent investment in wind turbine, GEG faces the risk of cash drainage, which may lead to insolvency. We expect PNJ to alleviate this threat through a strong cash flow, healthy balance sheet, a lower amount of long-term debt and support from its parent group-TTC .

APPENDIX NETWORK

APPENDIX A: FINANCIALS

Appendix A.1 – Income statement 8
Appendix A.2 – Balance sheet 9
Appendix A.3 – Cash flow statement 10
Appendix A.4 – Comparative income statement 11
Appendix A.4 – Comparative balance sheet 12
Appendix A.5 – Key financial ratios 13
Appendix A.6 – Altman's EM Z-score 13

APPENDIX B: BUSINESS

Appendix B.1 – Company structure 14
Appendix B.2 – GEG's subsidiaries 14
Appendix B.3 – GEG's hydroelectric dams 15
Appendix B.4 – GEG upcoming wind power plant project 15
Appendix B.5 – GEG upcoming solar power plant project 15

APPENDIX C: INDUSTRY AND COMPETITION

Appendix C.1 – Renewable electricity value chain 16
Appendix C.2 – Porter's Five Forces analysis 16
Appendix C.3 – Vietnam power development plan 17
Appendix C.4 – Competitive market phases 17

APPENDIX D: VALUATION

Appendix D.1 – Income statement assumptions 17
Appendix D.2 – Balance sheet assumptions 18
Appendix D.3 – Forward segment revenue 19
Appendix D.4 – Forward total revenue 2022-2027 19
Appendix D.5 – Gross profit margin comparision 2021 19
Appendix D.6 – D/A comparision 2021 19
Appendix D.7 – Relative valuation 20
Appendix D.8 – Sensitivity valuation 20
Appendix D.6 – Recommendations 20

APPENDIX E: CORPORATE GOVERNANCE

Appendix E.1 – BOD structure 21
Appendix E.2 – ESG assessment in GEG 21

Order	Abbreviation	Full term
1	A/E	Asset/Equity
2	APAC	Asia Pacific Accreditation Cooperation
3	BOD	Board of Director
4	GPM	Gross Profit Margin
5	CAGR	Compounded Annual Growth Rate
6	Capex	Capital Expenditure
7	COD	Commercial operation day
8	CSR	Corporate Social Responsibility
9	D/A	Debt/Asset
10	DPPA	Direct Power Purchasing Agreement
11	EVN	Vietnam Electricity
12	FCFF	Free Cashflow to Firm
13	FDI	Foreign Direct Investment
14	FiT	Feed-in Tariff
15	GRI	Global Reporting Initiative
16	HOSE	Ho Chi Minh Stock Exchange
17	IFC	International Finance Corporation
18	IP	Industrial Park
19	RE	Renewable Energy
20	P/E	Price to Earning per Share
22	PE	Private Equity
23	ROE	Return on Equity
24	WACC	Weighted average cost of capital

APPENDIX A: FINANCIALS

APPENDIX A.1 – INCOME STATEMENT

	2016A	2017A	2018A	2019A	2020A	2021F	2022F	2023F	2024F	2025F
Net Sales	**431.4**	**539.1**	**559.5**	**1159.4**	**1494.5**	**1449.6**	**3181.7**	**3283.8**	**3289.5**	**3299.3**
Cost of goods sold	(212.5)	(227.3)	(253.5)	(509.7)	(732.7)	(628.7)	(1676.8)	(1700.4)	(1722.8)	(1746.1)
Gross profit	**218.9**	**311.8**	**306.0**	**649.7**	**761.8**	**820.8**	**1504.9**	**1583.4**	**1566.7**	**1553.1**
Financial income	26.8	18.0	19.5	24.2	30.5	36.8	70.7	74.4	77.0	75.1
Financial Expenses	(45.8)	(24.6)	(41.1)	(257.8)	(371.2)	(343.6)	(834.6)	(805.6)	(775.4)	(745.3)
Of which: Interest Expenses	(39.1)	(24.4)	(40.2)	(244.9)	(354.4)	(327.3)	(798.9)	(768.6)	(738.4)	(708.2)
Selling Expenses	(0.1)	(1.0)	(0.6)	(0.6)	(1.7)	(1.3)	(2.7)	(3.2)	(3.0)	(3.0)
General and admin expenses	(51.3)	(65.1)	(76.6)	(111.2)	(114.3)	(149.5)	(292.2)	(297.1)	(312.9)	(305.1)
Net Profit/Loss from Operating activities	**148.5**	**239.2**	**207.1**	**304.3**	**303.9**	**363.2**	**446.0**	**552.0**	**552.4**	**574.9**
Other profits/(loss)	0.7	(8.7)	(0.2)	(0.3)	4.7	4.7	4.7	4.7	4.7	4.7
Profit or lost from join venture companies	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0
Total profit before tax	**149.2**	**230.4**	**206.9**	**303.9**	**308.6**	**367.9**	**450.7**	**556.7**	**557.1**	**579.6**
Current Corporate Income Tax Expenses	(13.1)	(21.4)	(20.6)	(13.7)	(14.1)	(27.6)	(31.6)	(39.0)	(25.1)	(29.0)
Deferred Income Taxes Expenses	1.8	(0.3)	(1.0)	(0.3)	(0.5)	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)
Net profit after tax	**134.4**	**209.3**	**187.3**	**290.5**	**295.0**	**339.7**	**418.6**	**517.2**	**531.5**	**550.0**
Benefits of Minitory shareholders	(39.3)	(55.5)	(42.6)	(39.5)	(37.6)	(39.9)	(39.0)	(38.8)	(39.2)	(39.0)
Net profit after tax attributable to parent company shareholders	**95.1**	**153.8**	**144.7**	**251.0**	**257.4**	**299.8**	**379.6**	**478.3**	**492.2**	**511.0**
Basic EPS	1.28	1.80	1.25	1.06	1.03	1.0	1.2	1.6	1.6	1.7
Diluted EPS				1.06	1.03					

APPENDIX A.2 – BALANCE SHEET

	2016A	2017A	2018A	2019A	2020A	2021F	2022F	2023F	2024F	2025F
TOTAL ASSET	**1551.7**	**1658.6**	**4360.6**	**6763.2**	**7773.1**	**16014**	**18524**	**18079**	**17686**	**16777**
Cash and Cash Equivalents	68	223	609.4	105.4	135.4	624.6	36.6	46.2	154.6	359.3
Short term financial investment	0	8.2	252	12	0	0.0	0.0	0.0	0.0	0.0
Account Receivables	257	227.4	245.5	688.2	1434.3	957.2	1941.4	1839.4	1677.7	1517.9
Inventory	16.2	26.2	24.7	314.3	152	231.6	406.7	463.1	528.5	526.6
Other Current Assets	0.7	1.1	98.2	70.5	57.6	75.4	67.8	67.0	70.1	68.3
Tangible Fixed Assets	1163.7	1107.4	2871.2	5239.7	5330.2	10111.5	14947.5	13998.5	13049.5	12100.5
Financial leases fixed assets	0	0	0	0	27.8	27.8	27.8	27.8	27.8	27.8
Intangible Fixed Asset	5.6	6.6	6.4	19.2	16.7	7.7	10.1	11.0	12.1	10.4
Construction in progress	5.6	18	215.8	262.8	541.4	3839.9	947.4	1487.4	2027.4	2027.4
Long term Finacial Investments	0.2	0	0	2.6	0	87.0	87.0	87.0	87.0	87.0
Other long-term assets	34	29.8	36.5	47.9	51.8	51.8	51.8	51.8	51.8	51.8
LIABILITIES	**401**	**340.6**	**1952.6**	**4183.6**	**4305.2**	**11874.6**	**13907.2**	**12837.7**	**11905.6**	**10436.6**
Account Payables	89.9	97.5	421.8	282.8	500.2	2500	2800.2	2271.4	1840.9	1492.7
Short-term borrowings	45.2	5.9	106.4	553.6	600.9	661.0	727.1	799.8	879.8	967.8
Other long-term payables	0.5	3.9	3.6	3.5	3.2	3.2	3.2	3.2	3.2	3.2
Long-term borrowings	264.5	232.4	1420.3	3343.1	3200.6	8672.3	10336.7	9721.4	9137.7	7926.8
Bonus and welfare funds	13.6	17.3	20.8	31.6	35.7	37.5	39.4	41.3	43.4	45.6
Technology-science development fund	0.9	0.9	0.6	0.6	0.3	0.6	0.6	0.6	0.6	0.6
OWNERS EQUITY	**1150.7**	**1318**	**2407.9**	**2579.6**	**3467.9**	**4139.8**	**4616.8**	**5241.3**	**5780.9**	**6340.3**
Common stock	744.9	970.9	1941.8	2038.9	2711.8	3037.0	3037.0	3037.0	3037.0	3037.0
Share premium	2	28.9	29	38.7	38.2	38.2	38.2	38.2	38.2	38.2
Other capitals of Owners Equity	0	0	0	0	0	0	0	0	0	0
Treasury stock	0	0	0	0	0	0	0	0	0	0
Development and Investment Fund	49.1	56.3	61	78.8	87.3	91.7	96.2	101.1	106.1	111.4
Retained earnings	168.4	120	145.1	164.8	220	560.0	978.7	1,496.1	2,027.3	2,577.3
Minority interest	186	142	231	259	411	413	415	418	420	423
Forecasting Discrepancies						0.06	51.51	151.42	152.13	153.66

APPENDIX A.3 – CASH FLOW STATEMENT

	2016A	2017A	2018A	2019A	2020A	2021F	2022F	2023F	2024F	2025F
Total profit before tax	**149.2**	**230.4**	**206.9**	**303.9**	**308.6**	**367.9**	**450.7**	**556.7**	**557.1**	**579.6**
Adjustments	-	-	-	-	-					
1. Depreciation of Fixed Assets Expenses	79.8	79.5	97.5	285.0	354.0	349.8	969.0	969.0	969.0	969.0
2. Provisions	-	0.3	0.3	0.5	(0.4)					
3. Unrealised foreign exchange gain/(loss)	-	-	0.8	0.5	1.4	1.4	1.4	1.4	1.4	1.4
4. Profit/(loss) from liquidating fixed assets	-	-	-	-	-					
5. Interest Expenses	39.1	24.4	40.2	247.3	357.9	327.3	798.9	768.6	738.4	708.2
6. Other deductions	0.9	-	-	-	-					
Operating profit/(loss) before changes in Working Capital	**242.0**	**323.5**	**326.6**	**813.5**	**995.7**	**1046.4**	**2220.0**	**2295.7**	**2265.9**	**2258.1**
Other adjustments	(11.2)	(18.1)	(95.2)	(295.7)	(667.3)	477.4	(666.1)	112.1	(61.8)	463.8
1. (Increase)/decrease in receivables	(5.9)	(10.0)	0.8	(289.4)	161.8	(79.5)	(344.0)	(29.4)	2.8	61.5
2. (Increase)/decrease in inventories	10.0	3.1	7.7	(282.4)	324.8	1999.8	300.4	(528.5)	(430.4)	(348.4)
3. Increase/(decrease) in payables	4.4	(0.4)	(97.1)	27.7	12.9	17.8	(7.6)	(0.9)	3.1	(1.8)
6. Interest paid	(42.6)	(31.3)	(50.1)	(246.7)	(356.2)	(253.3)	(817.9)	(800.6)	(777.4)	(733.2)
7. Income Tax paid	(19.4)	(16.9)	(25.8)	(13.0)	(11.3)	(27.6)	(31.6)	(39.0)	(25.1)	(29.0)
8. Other Incomes from Operating Activities	-	-	-	-	-					
9. Other Expenses from Operating Activities	(6.4)	(10.9)	(12.3)	(5.9)	(18.4)					
Net cashflow from operating activities	**168.4**	**243.8**	**141.1**	**(332.8)**	**420.6**	**3181.0**	**653.3**	**1009.5**	**977.1**	**1671.1**
1. Purchases of fixed assets and other long-term assets	(40.5)	(46.8)	(1810.0)	(2667.7)	(779.3)	(8389.5)	(2872.5)	(520.0)	(520.0)	20.0
2. Proceeds from disposal of fixed assets	0.9	2.7	0.2	2.8	32.1					
3. Changes in ST-LT Financial Investments	0.7	(8.0)	(243.8)	237.4	14.6	(87.0)	0.0	0.0	0.0	0.0
Interest, dividends, shared profit from investments	15.7	21.4	13.9	22.7	20.4	14.0				
4. Changes in other LT assets	114,1	35,3	93.8	(237.3)	(225.0)					
Net Cash flow from Investing activities	**90.9**	**4.6**	**(1.945.9)**	**(2.642.1)**	**(937.2)**	**(8462.5)**	**(2872.5)**	**(520.0)**	**(520.0)**	**20.0**
Proceeds from issue of shares	-	1.0	1.049.8	270.3	673.3	325.2	0.0	0.0	0.0	0.0
Payments for share returns and repurchases	-	-	-	-	-					
Increase / (Pay-off) of ST loans	(57.3)	(39.3)	100.5	447.2	47.3	60.1	66.1	72.7	80.0	88.0
Increase / (Pay-off) of LT loans	(253.0)	(32.1)	1187.9	1922.8	(142.5)	5471.7	1664.4	(615.3)	(583.7)	(1210.9)
Dividends paid	(39.5)	(29.5)	(150.7)	(166.9)	(35.6)	(86.2)	0.0	0.0	0.0	0.0
Changes in other LT liabilities	(1.9)	(9.7)	(8.9)	(8.2)	(7.3)					
Other distribution from equity	71.7	16.2	12.6	5.9	11.5					
Net Cashflow from Financing activities	**(280.0)**	**(93.4)**	**2.191.2**	**2.471.1**	**546.7**	**5770.8**	**1730.5**	**(542.6)**	**(503.7)**	**(1122.9)**
Net Cash and Cash equivalents during the period	(20.7)	155.0	386.4	(503.7)	30.1	489.3	(488.7)	(53.1)	(46.6)	568.2
Cash and cash equivalents at beginning of year	88.7	68.0	223.0	609.4	105.4	135.4	674.6	136.6	46.2	154.6
Cash and Cash equivalents at end of year	**68.0**	**223.0**	**609.4**	**105.4**	**135.4**	**624.6**	**36.6**	**46.2**	**154.6**	**359.3**

APPENDIX A.4 – COMPARATIVE INCOME STATEMENT

	2016A	2017A	2018A	2019A	2020A	2021A	2022F	2023F	2024F	2025F
Net Sales	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Cost of goods sold	-49.3%	-42.2%	-45.3%	-44.0%	-49.0%	-43.4%	-52.7%	-51.8%	-52.4%	-52.9%
Gross profit	**50.7%**	**57.8%**	**54.7%**	**56.0%**	**51.0%**	**56.6%**	**47.3%**	**48.2%**	**47.6%**	**47.1%**
Financial income	6.2%	3.3%	3.5%	2.1%	2.0%	2.5%	2.2%	2.3%	2.3%	2.3%
Financial Expenses	-10.6%	-4.6%	-7.3%	-22.2%	-24.8%	-23.7%	-26.2%	-24.5%	-23.6%	-22.6%
Of which: Interest Expenses	-9.1%	-4.5%	-7.2%	-21.1%	-23.7%	-22.6%	-25.1%	-23.4%	-22.4%	-21.5%
Selling Expenses	0.0%	-0.2%	-0.1%	-0.1%	-0.1%	-0.1%	-0.1%	-0.1%	-0.1%	-0.1%
General and admin expenses	-11.9%	-12.1%	-13.7%	-9.6%	-7.6%	-10.3%	-9.2%	-9.0%	-9.5%	-9.2%
Net Profit/Loss from Operating activities	**34.4%**	**44.4%**	**37.0%**	**26.2%**	**20.3%**	**25.1%**	**14.0%**	**16.8%**	**16.8%**	**17.4%**
Other profits/(loss)	0.2%	-1.6%	0.0%	0.0%	0.3%	0.3%	0.1%	0.1%	0.1%	0.1%
Profit or lost from join venture companies	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Total profit before tax	**34.6%**	**42.7%**	**37.0%**	**26.2%**	**20.6%**	**25.4%**	**14.2%**	**17.0%**	**16.9%**	**17.6%**
Current Corporate Income Tax Expenses	-3.0%	-4.0%	-3.7%	-1.2%	-0.9%	-1.9%	-1.0%	-1.2%	-0.8%	-0.9%
Deferred Income Taxes Expenses	0.4%	-0.1%	-0.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Net profit after tax	**31.2%**	**38.8%**	**33.5%**	**25.1%**	**19.7%**	**23.4%**	**13.2%**	**15.7%**	**16.2%**	**16.7%**
Benefits of Minitory shareholders	-9.1%	-10.3%	-7.6%	-3.4%	-2.5%	-2.8%	-1.2%	-1.2%	-1.2%	-1.2%
Net profit after tax attributable to parent company shareholders	22.0%	28.5%	25.9%	21.6%	17.2%	20.7%	11.9%	14.6%	15.0%	15.5%

APPENDIX A.5 – COMPARATIVE BALANCE SHEET

	2016A	2017A	2018A	2019A	2020A	2021F	2022F	2023F	2024F	2025F
TOTAL ASSET	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Cash and Cash Equivalents	4.4%	13.4%	14.0%	1.6%	1.7%	3.9%	0.2%	0.3%	0.9%	2.1%
Short term financial investment	0.0%	0.5%	5.8%	0.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Account Receivables	16.6%	13.7%	5.6%	10.2%	18.5%	6.0%	10.5%	10.2%	9.5%	9.0%
Inventory	1.0%	1.6%	0.6%	4.6%	2.0%	1.4%	2.2%	2.6%	3.0%	3.1%
Other Current Assets	0.0%	0.1%	2.3%	1.0%	0.7%	0.5%	0.4%	0.4%	0.4%	0.4%
Tangible Fixed Assets	75.0%	66.8%	65.8%	77.5%	68.6%	63.1%	80.7%	77.4%	73.8%	72.1%
Financial leases fixed assets	0.0%	0.0%	0.0%	0.0%	0.4%	0.2%	0.2%	0.2%	0.2%	0.2%
Intangible Fixed Asset	0.4%	0.4%	0.1%	0.3%	0.2%	0.0%	0.1%	0.1%	0.1%	0.1%
Construction in progress	0.4%	1.1%	4.9%	3.9%	7.0%	24.0%	5.1%	8.2%	11.5%	12.1%
Long term Finacial Investments	0.0%	0.0%	0.0%	0.0%	0.0%	0.5%	0.5%	0.5%	0.5%	0.5%
Other long-term assets	2.2%	1.8%	0.8%	0.7%	0.7%	0.3%	0.3%	0.3%	0.3%	0.3%
LIABILITIES	25.8%	20.5%	44.8%	61.9%	55.4%					
Account Payables	5.8%	5.9%	9.7%	4.2%	6.4%	74.1%	75.1%	71.0%	67.3%	62.2%
Short-term borrowings	2.9%	0.4%	2.4%	8.2%	7.7%	15.6%	15.1%	12.6%	10.4%	8.9%
Other long-term payables	0.0%	0.2%	0.1%	0.1%	0.0%	4.1%	3.9%	4.4%	5.0%	5.8%
Long-term borrowings	17.0%	14.0%	32.6%	49.4%	41.2%	0.0%	0.0%	0.0%	0.0%	0.0%
Bonus and welfare funds	0.9%	1.0%	0.5%	0.5%	0.5%	54.2%	55.8%	53.8%	51.7%	47.2%
Technology-science development fund	0.1%	0.1%	0.0%	0.0%	0.0%	0.2%	0.2%	0.2%	0.2%	0.3%
OWNERS EQUITY	74.2%	79.5%	55.2%	38.1%	44.6%	0.0%	0.0%	0.0%	0.0%	0.0%
Common stock	48.0%	58.5%	44.5%	30.1%	34.9%					
Share premium	0.1%	1.7%	0.7%	0.6%	0.5%	25.9%	24.9%	29.0%	32.7%	37.8%
Other capitals of Owners Equity	0.0%	0.0%	0.0%	0.0%	0.0%	19.0%	16.4%	16.8%	17.2%	18.1%
Treasury stock	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%	0.2%	0.2%	0.2%
Development and Investment Fund	3.2%	3.4%	1.4%	1.2%	1.1%	0.0%	0.0%	0.0%	0.0%	0.0%
Retained earnings	10.9%	7.2%	3.3%	2.4%	2.8%	0.0%	0.0%	0.0%	0.0%	0.0%
Minority interest	12.0%	8.5%	5.3%	3.8%	5.3%	0.6%	0.5%	0.6%	0.6%	0.7%

APPENDIX A.6 – KEY FINANCIAL RATIOS

	2016A	2017A	2018A	2019A	2020A	2021F	2022F	2023F	2024F	2025F
Profitability ratio										
Gross margin	50.74%	57.84%	54.69%	56.04%	50.97%	56.63%	47.30%	48.22%	47.62%	47.08%
Profit margin	31.15%	38.82%	33.48%	25.06%	19.74%	23.45%	13.16%	15.75%	16.15%	16.67%
ROE DuPont	11.68%	16.41%	11.27%	13.69%	9.10%	11.06%	9.73%	9.75%	9.09%	8.45%
ROA DuPont	8.66%	13.04%	6.22%	5.22%	4.06%	2.86%	2.42%	2.83%	2.97%	3.19%
EBIT/ Sales	43.65%	47.26%	44.16%	47.33%	44.36%	47.97%	39.27%	40.36%	39.38%	39.03%
NI/EBT	90.08%	90.84%	90.53%	95.59%	95.59%	92.34%	92.88%	92.90%	95.40%	94.90%
EBT / EBIT	100.40%	95.92%	99.61%	99.67%	100.98%	100.93%	100.44%	100.27%	100.31%	100.29%
Assets turnover	27.80%	33.59%	18.59%	20.85%	20.56%	12.19%	18.43%	17.94%	18.39%	19.15%
A/E	134.85%	125.84%	181.10%	262.18%	224.14%	386.84%	401.23%	344.93%	305.95%	264.61%
Efficiency ratio										
Days of sales outstanding (DSO)	266.20	161.74	152.14	144.96	255.84	237.64	219.64	201.64	183.64	165.64
Days of inventory on hand (DOH)	27.44	33.58	36.14	119.72	114.55	132.55	87.31	98.06	110.44	108.58
Days payable outstanding (DPO)	154.93	148.40	368.73	248.83	192.36	858.75	601.12	480.90	384.72	307.78
Cash conversion cycle	138.72	46.91	-180.45	15.85	178.04	-488.56	-294.18	-181.21	-90.64	-33.56
COGS / Inventory	13.12	10.72	9.96	3.01	3.14	3.28	5.25	3.91	3.47	3.31
Liquidity ratio										
Current ratio	2.30	4.03	2.24	1.37	1.57	0.59	0.69	0.78	0.88	0.99
Quick ratio	2.19	3.80	2.02	0.93	1.38	0.49	0.55	0.61	0.66	0.75
Cash Ratio	0.46	1.85	1.11	0.12	0.12	0.20	0.01	0.01	0.06	0.14
Debt / Asser	0.26	0.21	0.45	0.62	0.55	0.74	0.75	0.71	0.67	0.62
Debt / Equity	0.35	0.26	0.81	1.62	1.24	2.87	3.01	2.45	2.06	1.65
Short-term debt / Equity	0.13	0.09	0.23	0.34	0.33	0.77	0.77	0.59	0.48	0.40
Long-term debt / Equity	0.23	0.18	0.59	1.30	0.92	2.10	2.24	1.86	1.58	1.25
Interest coverage ratio	4.80	10.80	6.15	2.24	1.86	2.11	1.56	1.72	1.75	1.81

APPENDIX A.7 – ALTMAN Z-SCORE MODEL

The Z Score is a statistical measure of the distance (measured by standard deviation) from any data point to the mean of a data set. In finance. the Z-score is a credit test result that measures the likelihood of bankruptcy.

Z score = 1.2*A1 + 1.4*A2 + 3.3*A3 + 0.6*A4 + 0.999*A5

In which:

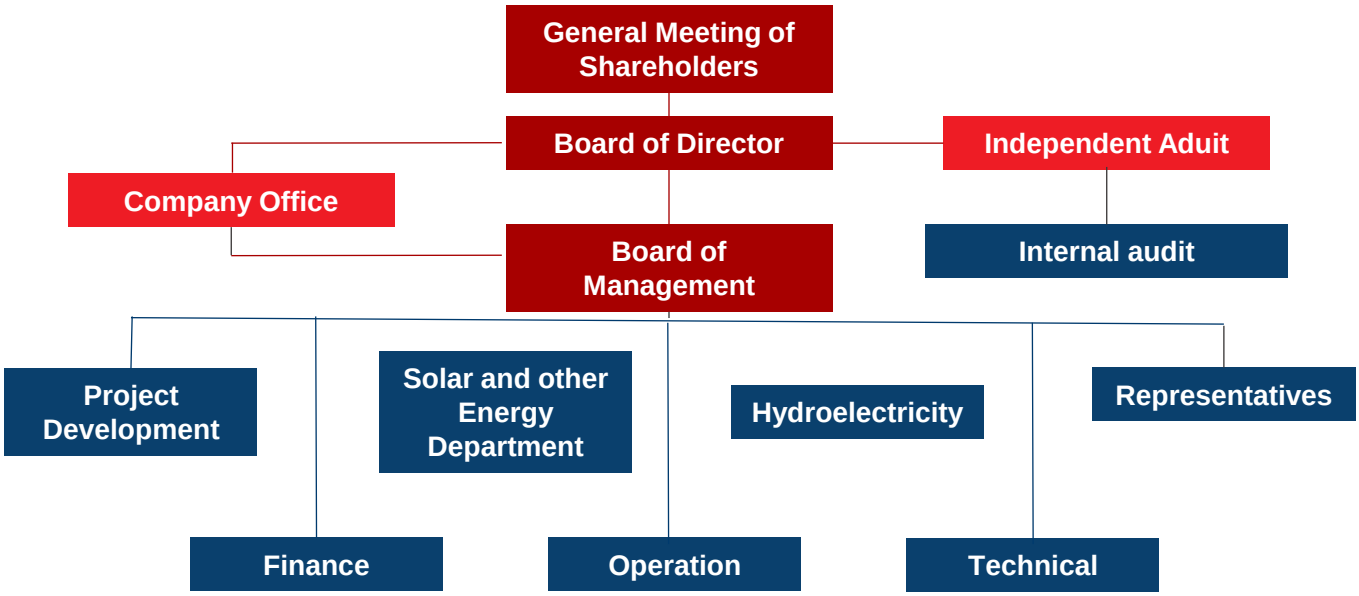
- 2.99 < Z: Low bankruptcy probability
- 1.81 < Z < 2.99: Moderate bankruptcy probability
- Z < 1.81: High bankruptcy probability

Input	2016	2017	2018	2019	2020
Revenue	431.4	539.1	559.5	1159.4	1494.5
Working Capital	206.8	382.5	701.6	354	678.2
Undistributed Net Profit	168.4	120	145.1	164.8	220
EBIT	187.6	263.6	247.3	549.2	658.3
Stock price		9.95	11.65	20.42	18.2
Free-float		970.9	1941.8	2038.9	2711.8
Total debt		340.6	1952.6	4183.6	4305.2
Total asset		1658.6	4360.6	6763.2	7773.1

Criteria	2017	2018	2019	2020
X1 (Working capital/Asset)	0.231	0.161	0.052	0.087
X2 (Retained Earning/Asset)	0.072	0.033	0.024	0.028
X3 (EBIT/Asset)	0.159	0.057	0.081	0.085
X4 (Market Cap/Debt)	28.363	11.586	9.952	11.464
X5 (Revenue/asset)	0.325	0.128	0.171	0.192
Z-score	**19.382**	**7.971**	**6.906**	**7.954**

APPENDIX B: BUSINESS

APPENDIX B.1 – COMPANY STRUCTURE



APPENDIX B.2 – GEG'S SUBSIDIARIES

Subsidiaries	Headquater	Business sector
1. Gia Lai hydroelectricity, JSC. (i)	Gia Lai	Electricity production and distribution
2. Thuong Lo hydroelectricity, Ltd.	Thua Thien Hue	Electricity production and distribution
3 . Gia Lai Construction, Ltd	Gia Lai	Electricity Construction
4. TTC Tay Son-Binh Dinh, JSC	Binh Dinh	Electricity production and distribution
5 . TTC Solar Power Tay Son-Binh Dinh, JSC	Binh Dinh	Electricity production and distribution
6. TTC Đuc Hue - Long An, JSC (ii)	Long An	Electricity production and distribution
7 .TTC Solar Power Đuc Hue - Long An, JSC	Long An	Electricity production and distribution
8. Gia Lai Development, Ltd.	Gia Lai	Electricity Construction
9. Truc Son Solar Power Plant, Ltd. (ii)	Dak Nong	Electricity production and distribution
10. VPL Energy, JSC. (iii)	Ben Tre	Electricity production and distribution
11. Tien Giang Wind Power, JSC. (iv)	Tien Giang	Electricity production and distribution
12. Ia Bang Wind Power, JSC. (v)	Gia Lai	Electricity production and distribution
13. Vi-Ja Renewable Energy, JSC. (vi)	TP.HCM	Electricity production and distribution
14. Bac-Ayun Ha, JSC. (vii)	Gia Lai	Electricity production and distribution
15. Phong Dien Solar Power, JSC.	Thua Thien Hue	Electricity production and distribution
16. TTC Bac Ninh-Bình Thuan, JSC.	Binh Thuan	Electricity production and distribution
17. TTC Solar Power Bac Ninh-Binh Thuan, JSC.	Binh Thuan	Electricity production and distribution
18. TTC Energy Bac Ninh-Binh Thuan	Binh Thuan	Electricity production and distribution

APPENDIX B.3 – GEG'S HYDROELECTRIC DAMS

Dam	Region	Water source	Year into operation	Capacity (MW)	Reservoir (million m3)	Average Output (million kWh/year)
H'Mun	Gia Lai	Ayun River	2010	16.2	0.87	67
Ayun Thượng		Ayun River	2011	12	4.54	51
H'Chan		Ayun River	2006	12	0.77	57
Dak Pi Hao 2		Dăk Pi Hao Stream	2008	10	0.2	37
Da Khai	Lam Dong	Da Nhim River	2010	8.1	9.95	37
la Puch 3	Gia Lai	Puch Stream	2009	6.6	0.12	31
Thuong Lo	Hue	Ba Ran River	2015	6	4.03	18
Dak Pi Hao 1	Gia Lai	Dak Pi Hao Stream	2014	5	0	21
Ayun Ha		Ayun River	2001	3	253	20
la Meur 3		Meur Stream	2005	1.8	0.02	8
la Drang 3		Drang Stream	2005	1.6	0.4	9
la Drang 2		la Drang Stream	1989	1.2	0.01	7
la Drang 1		la Drang Stream	2002	0.6	0.05	4
Total				**84.1**	**253**	**367**

APPENDIX B.4 – GEG UPCOMING WIND POWER PLANT PROJECTS

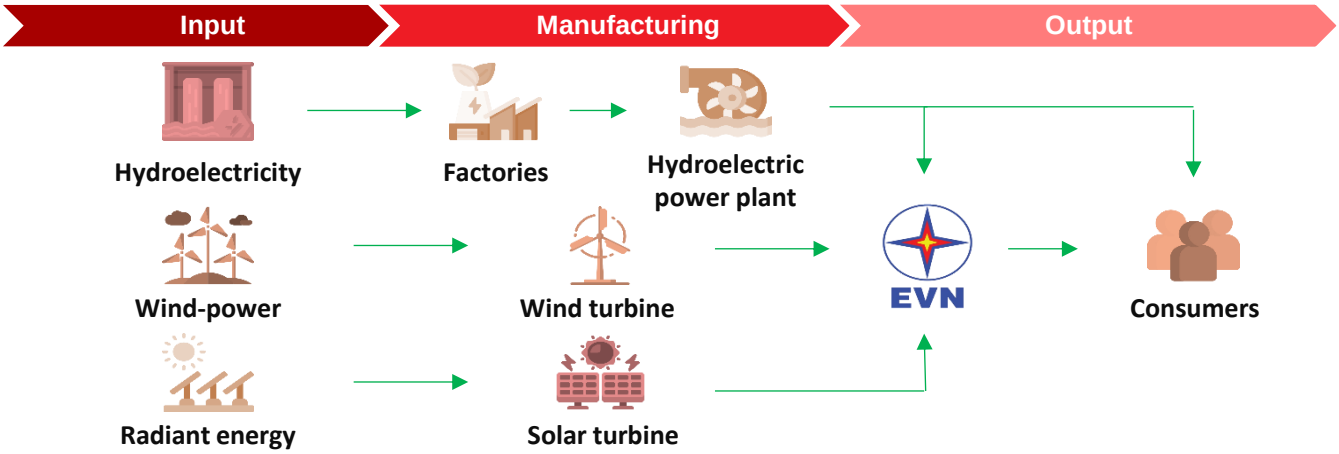
Plant	Investment	Capacity	Type	Province	Wind-speed	FIT	Year into operation
Tan Phu Dong 1	~VND 4 trillion	100MW	Offshore	Tien Giang	6.23 m/s	Unannounced	2022
VPL 2	~VND 1	30MW	Offshore	Ben Tre	6.35 m/s	Unannounced	2023

APPENDIX B.5 – GEG UPCOMING SOLAR POWER PLANT PROJECTS

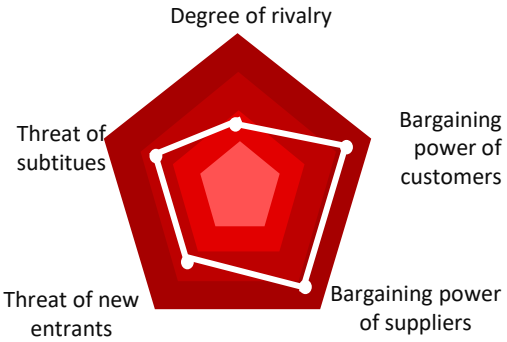
Plant	Investment	Capacity	Province	Radiation	Sunny hours	Year into operation	FIT
Duc Hue 2	~ VND 950 million	49MWp	Long An	4.9 kwh/m2/day	2445 hours/year	2023	Unannounced

APPENDIX C: INDUSTRY AND COMPETITION

APPENDIX C.1 – RENWEABLE ELECTRICITY VALUE CHAIN

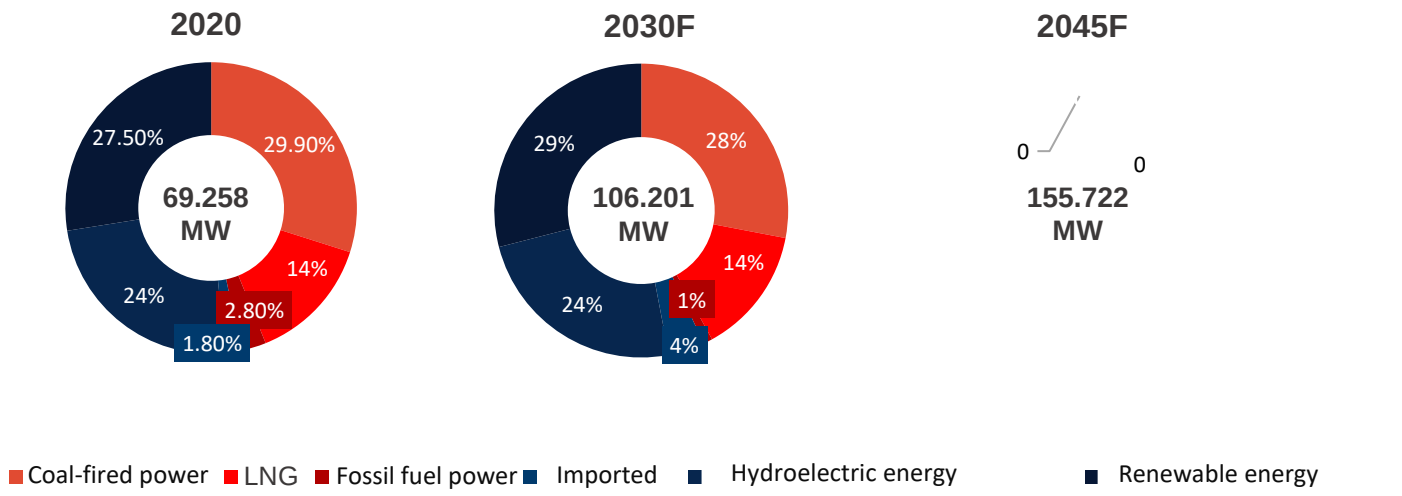


APPENDIX C.2 – PORTER'S FIVE FORCES ANALYSIS



Porter's Force	Impact	Description
Bargaining power of customers	High	High bargaining power of customers: The only customers of electricity producers is EVN, thus, GEG cannot choose other buyers rather than EVN
Degree of rivalry	Low	Low competition from rivals: • Renewable projects are subjected to the regulations and the source of fund. GEG has all of its operated at the highest FiT price and received long-term cooperation from renowned bank to finance future projects. • GEG's power plants in Gia Lai, Tien Giang, Ben Tre makes its possible for the firm to avoid peak-season capacity cutdown from EVN. In the other hands, its rival must follow EVN schedule to cut down its capacity of 10%-20% during peak season.
Bargaining power of suppliers	High	High bargaining power from suppliers: Due to the high-quality requirements of the industry, the number of suppliers that meet the condition to sell turbine, solar battery, electricity devices, … is relatively low. Thus, suppliers have stronger ability to raise equipment price.
Threat of new entrants	Moderate	Moderate threat due to the following factors: • More and more renewable energy developers are joining the industry thanks to the generous support from the government's regulations. • High CAPEX and underdeveloped infrastructure hinders the number of new entrants.
Threat of subtitutes	Moderate	Moderate threat of subtitutes: Although, there are lots of other alternative power such as: biomass, LNG, nuclear,… that bear the same characteristics as wind-solar-hydro energy but those kind of energy are considered dangerous for the humankind and Vietnam is currently incapable of managing them.

APPENDIX C.3 – VIETNAM POWER DEVELOPMENT PLAN



Source: Ministry of Industry and Trade

APPENDIX C.4 – COMPETITVE MARKET PHASES

APPENDIX D: VALUATION

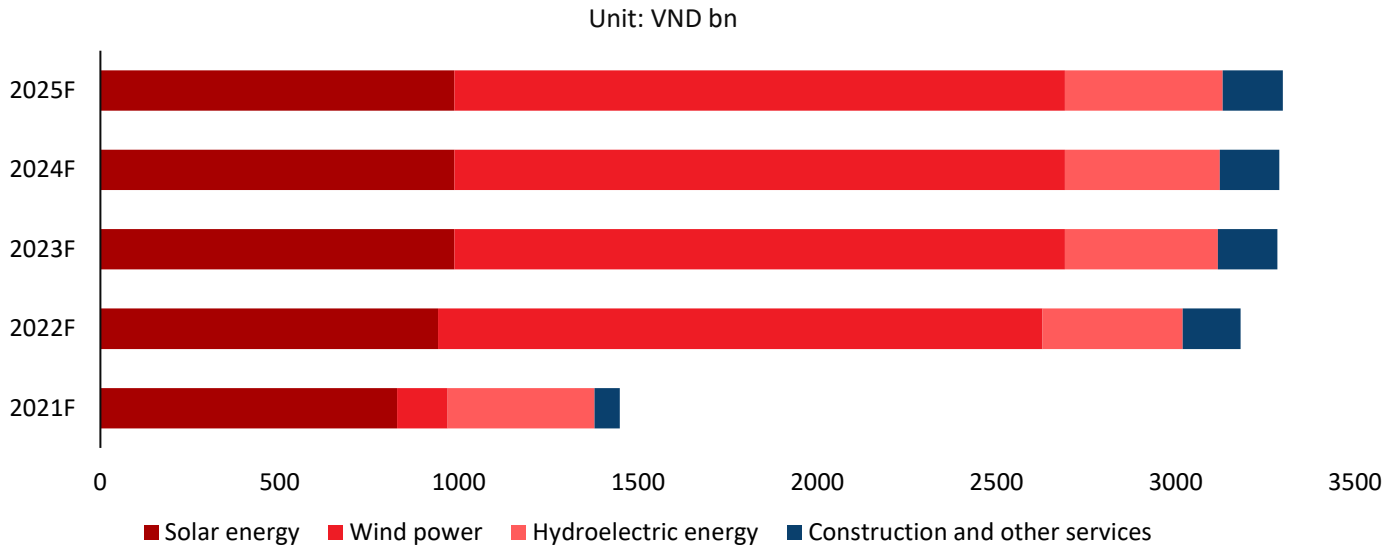
APPENDIX D.1 – BALANCE SHEET ASSUMPTIONS

Items	Assumptions
Day Receivable/Revenue	5% reduction compared to 2020 due to the implementation of TPD1, VPL2 projects by enterprises. 15% off after that
DOH/Revenue	Strong increase in 2021 (according to the company's expected data) and decrease by 20% after that
Long-term Debt	Besides assuming the company will gradually repay its debt in the future, we believe that the company will add about VND 2 trillion in long-term debt in 2022 to finance 2 wind power projects VPL2 and Tan Phu Dong 1. in Tien Giang (Total investment is about 5 trillion, 70% financed by debt, about 3.5 trillion, in 2021, 1.5 trillion has been disbursed)
Short-term Debt	Increase 10% each year
%Cash Dividend	GEC is in the investment phase so we expect the cash dividend to drop to 3% per year

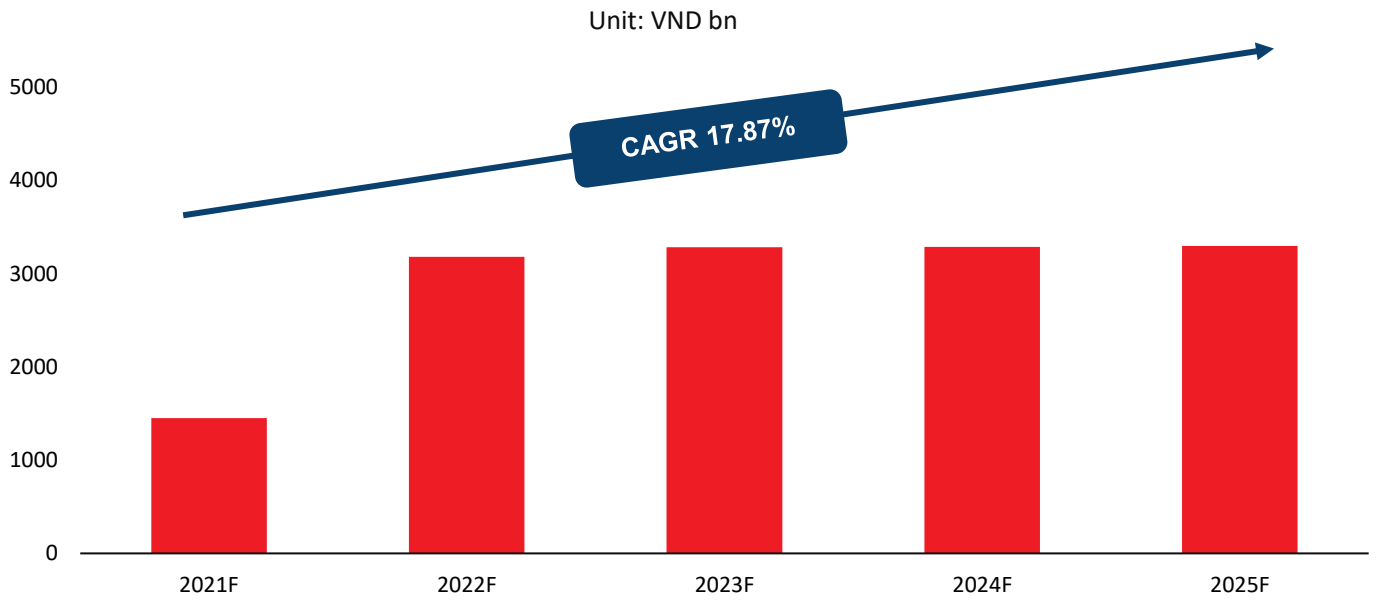
APPENDIX D.2 –INCOME STATEMENT ASSUMPTIONS

Items	Income Statement Assumptions
Solar power	Solar radiation falling on Vietnam is forecasted to increase sharply in the coming years, especially in the Central and Southern regions. GEC is developing a team of experienced technicians to run factories better. EVN will upgrade the power transmission line to meet the electricity output according to Circular 55. Therefore, we forecast that the actual output divided by theory (Qr/Q) of the solar power array will be **105% from next year**. 2023. Duc Hue 1-Long An solar power plant alone will be **115% from 2023** because the demand for electricity in the South and heat radiation in this area will increase sharply. In the two years 2021 and 2022, due to the complicated situation of the Covid epidemic, we have set the Qr/Q level at **90% and 100% respectively.**
Wind power	In 2021, 3 wind power plants will officially come into operation and enjoy preferential FIT prices: Ia Bang 1, Tan Phu Dong 2 and VPL1-Ben Tre. In which, Ia Bang 1 is an onshore project, so the **FIT price is 8.5 cents/kwh**; Tan Phu Dong 2 and VPL1 are nearshore projects, so the **FIT price 9.8 iscents/kwh.** We forecast that the output of these 3 factories will be **16% compared to the average output in 2021** (Since these factories will come into operation in Q4/2021 when the epidemic situation is under control, industrial zones are gradually reopening, leading to a recovery in electricity demand) and **100% for the following years** due to GEC's improvement in operating quality, EVN's upgrading of power transmission lines and geographical location in the strong wind. The remaining 2 plants, VPL2-Tien Giang and Tan Phu Dong 1 (2 projects near the shore) are still waiting to finalize electricity prices. According to your management, the selling price of electricity is expected to **decrease by 10% compared to the nearshore FIT price**.
Hydroeclectricity	For the hydropower segment, we forecast to operate at a **Qr/Q level of 95%** from 2022 due to the return of La Nina phenomenon.
Construction	We forecast that revenue from this segment will **account for 5% in 2021** as the COVID-19 pandemic has reduced GEC's sales performance; and **maintained at 5%** for the years as planned by the Board of Directors and GEC is focusing on the renewable energy segment.
Gross Profit Margin	We expect gross profit margin to grow at a **CAGR of 13%** from 2021-2025. In which, gross profit margin will decrease to 47% in the period 2021-2022 due to the sharp increase in depreciation expense from new factories.
COGS	_We forecast material cost will **increase 2% per year** due to inflation. _ Labor costs in 2021 decreased sharply due to GEC's personnel restructuring. Previously, in 2019 when GEC carried out the construction of solar power plants, it also restructured once. That's why we set the 2021 labor cost at **10 VND/kwh**. The following years is **100 VND/kwh** _We estimate the cost of outsourcing will **decrease by 10% per year from 2021** as GEC is upgrading the in-house technician team to maintain the factories without having to outsource them.

APPENDIX D.3 – FORWARD SEGMENT REVENUE



APPENDIX D.3 – FORWARD SEGMENT REVENUE



APPENDIX D.4 – GROSS PROFIT MARGIN COMPARISION 2021

Ticker	GPM (%)	Net Margin (%)
GEG	58	22
PC1	10	8
TTA	58	22
BCG	38	37
LCG	21	10
GEX	15	6
REE	37	31
Average	**38**	**22**

APPENDIX D.5 – D/A COMPARISION 2021

Ticker	D/A
GEG	71.80%
PGV	76.20%
BCG	82%
GEX	64%
LCG	69%
PC1	66%
Average	**71.50%**

APPENDIX A.A – RELATIVE VALUATION

COMPANY	COUNTRY	Market capitalisation	ROE (TTM)	ROA (TTM)	P/B (Q)	P/E (TTM)
SAHACOGEN CHONBURI	Thailand	3286	9.3%	6.2%	10.8	26.8
GEPIC ENERGY	China	11292	6.60%	2.60%	14.3	19.5
Guizhou Qianyu	China	21787	14.30%	2.60%	3.4	15.4
Guangzhou Heng	China	20500	7.70%	3.10%	1.3	14.1
PetroVietnam Power Nhon Trach 2 JSC	Vietnam	7456	15.0%	9.1%	1.8	12.1
Vinh Son - Song Hinh Hydropower JSC	Vietnam	6803	10.20%	3.70%	1.8	19.1
Power Generation Joint Stock Corporation 2	Vietnam	36786	N/A	N/A	1.8	21.7
PetroVietnam Power Corporation	Vietnam	40046	9.6%	5.40%	1.2	14.7
MEDIAN		**15896**	**9.8%**	**3.7%**	**1.8**	**17.3**
Gia Lai Electricity JSC (GEG)	Vietnam	**5937**	**9.6%**	**3.1%**	**1.7**	**22.8**
EPS			1.12			
P/E			17.3			
Target Price (1000 VNĐ)			**19**			

Source: Refinitiv Eikon

APPENDIX A.A – SENSITIVITY VALUATION

		g				
	25.68	1.50%	1.75%	2%	2.25%	2.50%
WACC	8.48%	27.61	29.77	32.09	34.60	37.32
	8.73%	24.67	26.64	28.76	31.05	33.51
	8.98%	21.94	23.75	**25.68**	27.78	30.02
	9.23%	19.39	21.06	22.84	24.75	26.81
	9.48%	17.03	18.56	20.20	21.96	23.83

APPENDIX A.A – RECOMMENDATIONS

Rank	Detail
Buy	Upside >10%
Hold	Upside between 5% to 10%
Approriate	Upside between -10% to below 5%
Less Positive	Downside between -20% to below -10%
Sell	Downside more than 20%

APPENDIX E: CORPORATE GOVERNANCE

APPENDIX E.1 – BOD STRUCTURE

No	Name	Responsibilities	Stock	Ownership	Academic achievement
1	Tan Xuan Hien	Head of BOD	30466	0.01%	Bachelor in Electricity Engineering
2	Nguyen Thuy Van	Member of BOD	-	-	-
3	Deepak Chand Khanna	Member of BOD	-	-	MBA
4	Dang Huynh Tuan Anh	Member of BOD	-	-	-
5	Nguyen The Vinh	Member of BOD	-	-	MBA
6	Pham Hong Duong	Independent member of BOD	-	-	MBA
7	Phạm Thi Khue	Independent member of BOD	-	-	-
8	Nguyen Thai Ha	CEO	505911	0.17%	Master of economy
9	Le Thanh Vinh	Vice CEO	494403	0.16%	Master of Economy
10	Nguyen Phong Phu	Vice CEO	279250	0.09%	-
11	Tran Thi Hong Tham	CFO	149250	0.05%	-
12	Tran Thi Hong Tham	Chief Accountant	149250	0.05%	-
13	Pham Thanh Tuan Anh	Legal Representative	15003	-	-

APPENDIX A.A – ESG ASSESSMENT IN GEG

1. Environment

- GEG complies with International and National Standards on Environmental Protection such as: all projects meet the standards of Environmental and Social Impact Assessment (E&S) Report of IFC - International Monetary Fund; closely follow international standards such as the ASEAN Corporate Governance Scorecard Assessment 2011 and 2019, the Corporate Governance Code of the Organization for Economic Co-operation and Development (OECD 2015),... and especially the consolidated GRI Sustainable Development Reporting Standards 2016; regulations on Vietnam's Law on Environmental Protection, etc. Therefore, GEG's projects and operations have minimized impacts on the environment, society, community and ecosystem. around of the projects.

- GEG upholds the responsibility to protect the natural ecosystem. Specifically, the company has focused on surveying birds, bats and benthic fauna before, during the start of construction and after construction; Measure the noise in the area around the project to consider, predict and come up with appropriate solutions in order to minimize the impact of the project when investing on the ecosystem.

- GEG chose the solution to use the main products manufactured by the Company such as: inventing and using GEG Robotic to clean the battery to save water up to 10 times; Refining Hydraulic Oil for reuse should reduce the amount of Oil purchased significantly, reducing by 83% in 2020 compared to 2019. This not only helps GEG reduce input costs but also reduces the volume of hazardous waste generated. born; using electricity produced by GEG itself for daily activities,... This helps to save maximum water, electricity and reduce hazardous waste to the environment.

- All Plants mainly use materials available in nature such as Water, Wind, and Sunlight instead of other materials to produce Electricity. As a result, the Company significantly limits greenhouse gas emissions. During the five-year period 2016-2020, the Company's total CO2 deflation nearly tripled, with a compound annual growth rate of 18%, bringing the total CO2 emissions to 2,075,455 tons.

2. Social

- **Local Habitat:**

GEG has factories located in underdeveloped economic areas, remote areas, helping to create more jobs, improve infrastructure, and stimulate local economic growth.

GEG participates in voluntary activities such as donating to support the people in the Central region, organizing voluntary blood donation programs, supporting natural disasters in Phong Dien NM area, study promotion association... with the total value recorded. received in 2020 about 708 million VND. From 2012 to 2020, GEG has contributed about 13 billion VND to CSR - Corporate Social Responsibility activities.

- **Gender equality**: With the specificity of industries in the field of Energy, the percentage of female employees is low compared to male employees, accounting for about 20% in 2020. The percentage of female employees working in management accounts for 26% of the total number of employees. management: 2/5 members of the Board of Directors are female, accounting for 40%; The Board of Directors has 3/5 people who are female, accounting for 60%. Female employees are guaranteed maternity leave, male employees are entitled to at least 5 days off when their wives give birth.

- **Labor:**

In terms of salary, bonus and welfare regime for employees, GEG has accumulated salary and welfare spending increased within 5 years from 2015 (with accumulated salary growth of 16%, accumulated benefit expenditure increased by 24%).

Regarding occupational safety and health, GEG also strengthens direct interaction with employees by establishing a Health and Safety network and organizing inspection of Occupational Safety; organizing emulation movements, movements for occupational safety and hygiene and building a culture of occupational safety at the workplace;...

Regarding training and development opportunities, GEG continuously trains to improve knowledge for employees. The average number of training hours for female employees is 8.2 hours/person and 1.6 hours/person for male employees; The average number of training hours for Managers is 15.9 hours/person and 1.6 hours/person for employees in 2020

However, in 2020, GEG will restructure its personnel towards lean and selective; new recruitment of 55 people - a decrease of 30% and leave of 85 people - an increase of 304% compared to 2019. The number of people who quit their jobs under the age of 30 is 33%, from 30 to 50 is 58% and over 50 is 9 years old. %. In the context of Covid-19, the restructuring strategy is not really appropriate in terms of social standards.

3. Governance

- **Independent - transparent - effective leadership structure:** GEG currently has 5 members of the Board of Directors, of which 3/5 are independent - accounting for 60%, exceeding the level specified in Decree 155/2020/ND-CP in terms of the number of members. members from 3-5 and at least 1 independent. Non-executive members account for 100%, independent is 60%, this is a percentage that is completely guaranteed that no one or a group of Members can dominate decision-making and the interests of the Company are always higher than the interests of the Company. interests of individual colleges.

- **100% of the Board of Directors have over 15 years of experience and high expertise**: the Board of Directors has diverse professional competencies in the fields of Energy 20%, Finance 60%, Management - Executive 100%; 20% Foreigners; 40% Female; 60% Independent; 100% Non-Executive; 100% of BOD members are those with more than 15 years of experience in Administration - Executive in many fields including Energy, Management, Auditing, Finance.

- **Effective succession planning program:** GEG conducts the development of a plan to find and train potential personnel, a diverse succession of leaders in terms of expertise, experience, gender, age and engagement group, in line with the development strategy. for members of the Board of Directors, CEO and key Board of Directors to ensure growth and continuous increase in shareholder value.

- There is an Independent Audit Committee, the Internal Auditor complies with the information disclosure rules and the reporting system.

- Protecting interests, treating all shareholders fairly, including foreign shareholders.